Marwadi University Marwadi Chandarana Group NAAC A+	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim:-Simulate Hit or Miss Transformation on images.**	
Experiment:-12	**Date:31-10-2026**	**Enrollment No:92400133063**

Code:-

```
import cv2
import numpy as np
from google.colab.patches import cv2_imshow
# Load the binary image
image = cv2.imread('/content/images.jpeg', 0)

# Define the structuring element for the pattern to be detected (0 for don't care, 1
for foreground, -1 for background)
pattern = np.array([[-1, 1, -1],
                    [1, 1, 1],
                    [-1, 1, -1]], dtype=np.int8)

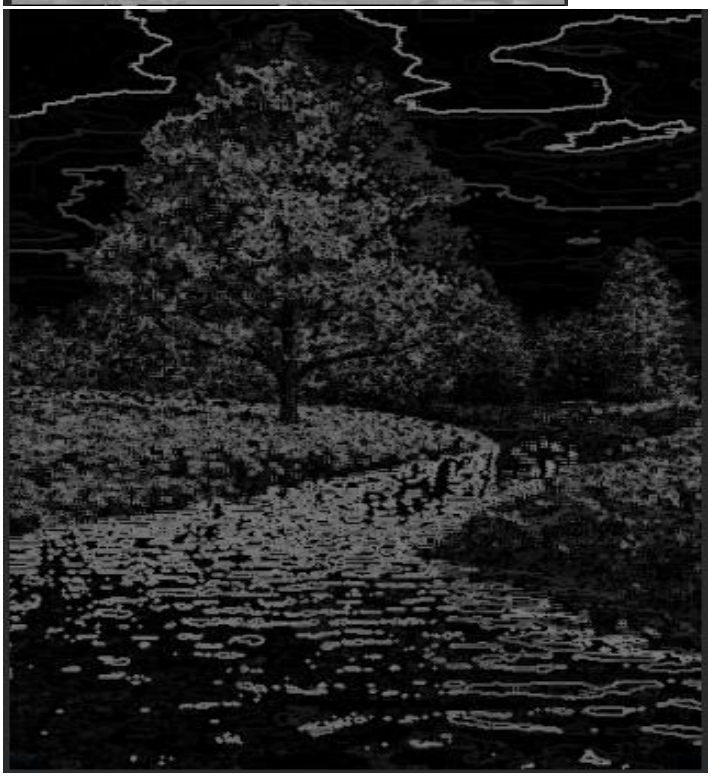
# Apply the Hit or Miss Transformation
result = cv2.morphologyEx(image, cv2.MORPH_HITMISS, pattern)

# Display the original image and the result
cv2_imshow(image)
cv2_imshow(result)
cv2.waitKey(0)
cv2.destroyAllWindows()
```

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output:-





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Conclusion:-

The given experiment explain the Hit-or-Miss Transform, a morphological operation used to detect specific spatial patterns (such as corners, junctions, or endpoints) in a binary image. The structuring element uses a ternary encoding—1 for foreground pixels that must match, -1 for background pixels that must match, and 0 for don't-care positions—enabling precise template matching at every pixel location. The result highlights only those pixels where the exact cross/plus-shaped pattern is found, making this technique valuable for feature detection, shape recognition, and thinning/skeletonization preprocessing in binary image analysis.